IVAN PUJOL VIDAL

Mechanical Engineer

PROFESSIONAL PROFILE

Mechanical Engineer at Alfa Laval with a Master's in Mechanical Engineering. Specialized in advanced CFD simulations using STAR-CCM+, with a Master's thesis focused on aerodynamic optimization and drag reduction. Experienced in industrial design, technical calculations, and quality systems (ISO 17025). Passionate about applying fluid dynamics and self-taught OpenFOAM skills to solve complex challenges in the process and life science sectors.

WORK EXPERIENCE

Alfa Laval (Mechanical Engineer)

09/2024 - Present (Copenhagen, Denmark)

I work in the Engineering & Supply division within the Fresh Water Generators business unit at Alfa Laval. My responsibilities include:

- Designing mechanical components, systems, and plants for industrial applications.
- Preparing technical specifications, datasheets, and project documents.
- Performing technical calculations and verifications to ensure design integrity.
- Engineering solutions based on customer needs, standards, and codes.
- Improving existing designs and supporting new technical developments.
- Attending internal and customer project and verification meetings.
- Coordinating with procurement to align with suppliers on technical matters.

Vadecity (Mechanical Design Engineer)

04/2023 - 07/2024 (Barcelona, Spain)

Vadecity, Spanish start-up in micromobility projects. Responsibilities:

- Evaluated products for optimization and industrialization, boosting efficiency.
- Utilized SolidWorks to create detailed 3D models.
- Utilized SolidWorks Simulation to perform Finite Element Analysis (FEA), ensuring robust and reliable product designs.
- Led prototyping efforts for rapid design iteration.
- Implemented designed solutions in major cities such as Barcelona and Madrid.

SEAT (Quality engineer intern)

09/2022 - 03/2023 (Martorell, Spain)

Implemented a Quality Management System based on the ISO 17025 standard.

EDUCATION

Master thesis student | Mechanical Engineering

09/2024 - 02/2025 (Copenhagen, Denmark)

Technical University of Denmark

Master's Degree in Industrial Engineering (Mechanical specialization)

09/2022 - 02/2025 (Barcelona, Spain)

Universitat Politècnica de Catalunya

MSc degree in Environmental Engineering modules

02/2022 - 07/2022 (Padova, Italy)

Universitá degli Studi di Padova (Exchange program)

Bachelor's Degree in Industrial Engineering

09/2018 - 01/2023 (Barcelona, Spain)

Universitat Politècnica de Catalunya

CONTACT

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SKILLS

CAD · Technical drawing · 3D Print
Prototyping · CFD · Finite Element
Analysis · Aerodynamics · Autocad
Solidworks · STAR CCM+ · ANSYS
Learning OpenFOAM (Self-study)

RELEVANT PROJECTS

Master thesis - CFD Simulations

Grade A - Master thesis on CFD simulations of the aerodynamics of speedway motorcycles using STAR CCM+, focused on analyzing drag forces and key drag elements. Conducted wind tunnel tests to validate CFD results and redesigned components to minimize drag. In collaboration with Team Denmark to apply findings to their motorcycles.

OTHER COURSES

Applied Computational Fluid Dynamics

Siemens - Online course (July 2024)

Sports and Building Aerodynamics

Eindhoven University of Technology
Online course (June 2024)

LANGUAGES

Catalan	Native
Spanish	Native
English	Proficient
Italian	Intermediate